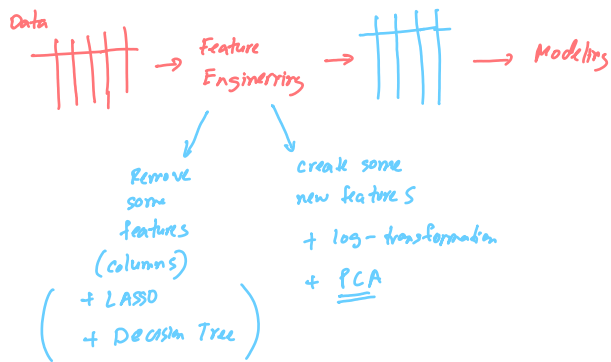




Principal Component Analysis

Son Nguyen

⑧



⑨

Information of data

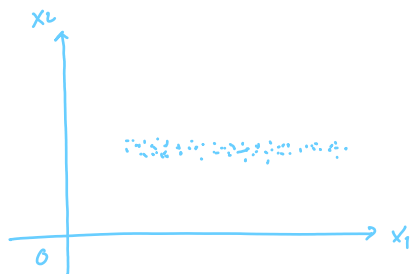
	Salary	GPA	Country	Gender	Expenses
	20	3.2	US	F	10
	90	3.21	China	F	10
	100	3.19	Can	F	10
	500	3.2	Mexico	F	10

Salary has more information than GPA

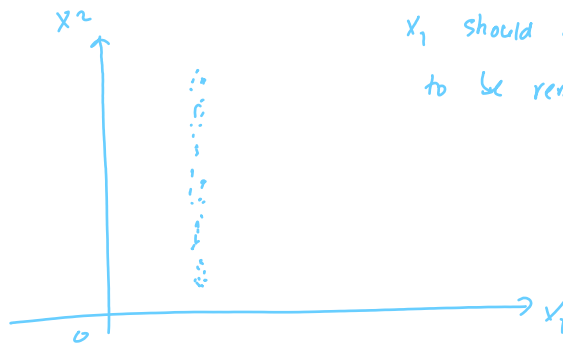
Country has more information than Gender

⇒ We can use the variance to measure the "information" of the variable.

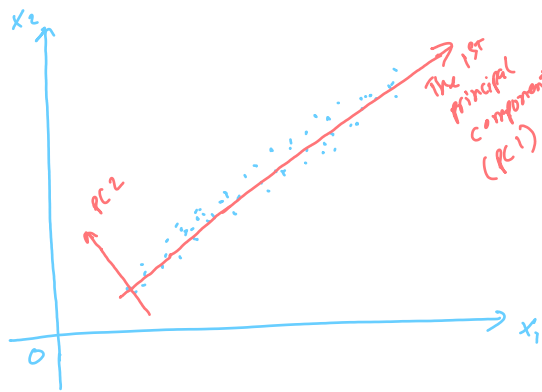
⑩



x_1 has way more "information" than x_2 and it contains almost all the information of the data. If we want to remove a variable, the variable should be x_2 .

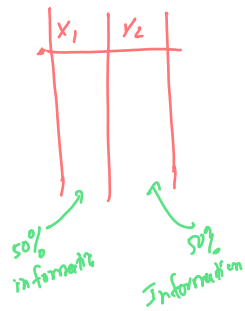


x_1 should be considered to be removed



We want to
create a new **variable**
that contains most
of the information of
the data

Original data

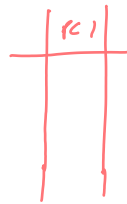


→



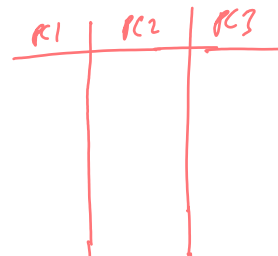
95% information (pointing to PC1)
5% information (pointing to PC2)

→ we remove PC2 →



40% info (pointing to x_1)
30% info (pointing to x_2)
30% info (pointing to y)

① →



85% (pointing to PC1)
10% (pointing to PC2)
5% (pointing to PC3)

(don't do this)

② →



60% (pointing to PC1)
10% (pointing to PC2)
30% (pointing to y)

(do this)

